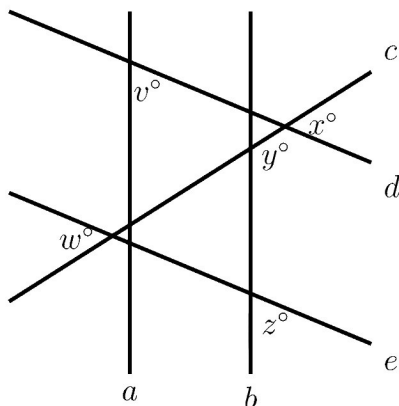


Lines&Angles

1. [June 2023]

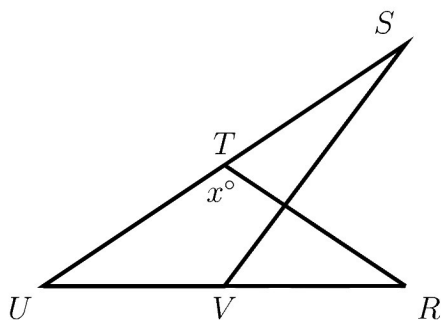


Note: Figure not drawn to scale

In the figure, parallel lines a and b are intersected by lines c , d , and e . If $x = 49$, $y = 136$, and $v < z$, which statement about x and w must be true?

- A) $x < w$
- B) $x > w$
- C) $x = w$
- D) $x + w = 90$

2. [November 2023]



In the figure, $RT = TU$, the measure of angle VST is 23° and the measure of angle RVS is 44° . What is the value of x ?

3. [March 2024]

A polygon has exactly 73 sides. If the measure of each of the 73 interior angles of this polygon is $(180p)^\circ$, what is the value of p ?

4. [August US 2023]

For two acute angles, $\angle A$ and $\angle B$, $\cos A = \sin B$. The measures, in degrees, of $\angle A$ and $\angle B$ are $x + 8$ and $4x + 2$, respectively. What is the value of x ?

- A) 37
- B) 34
- C) 16
- D) 2

5. [November 2023]

Quadrilaterals $PQRS$ and $WXYZ$ are similar, where P , Q , and R corresponds to W , X , and Y , respectively. The measure of $\angle S$ is 105° , $PS=25$, and $WZ=5$. What is the measure of $\angle Z$?

- A) 5°
- B) 21°
- C) 75°
- D) 105°

6. [December 2023]

For two acute angles $\angle Q$ and $\angle R$, $\cos Q = \sin R$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 21$ and $4x + 9$, respectively. What is the value of x ?

- A) 4
- B) 12
- C) 30
- D) 69

7. [March 2024]

Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(9x - 190)^\circ$. Which of the following could NOT be the sum of the measures of any two of these angles?

- A) $(-18x + 380)^\circ$
- B) $(-18x + 740)^\circ$
- C) $(18x - 380)^\circ$
- D) 180°

8. [March 2024]

The measure of angle R is $\frac{2\pi}{3}$ radians. The measure of angle T is $\frac{13\pi}{18}$ radians greater than the measure of angle R . What is the measure of angle T , in degrees?

- A) 120 C) 250
B) 130 D) 500

9. [March 2024]

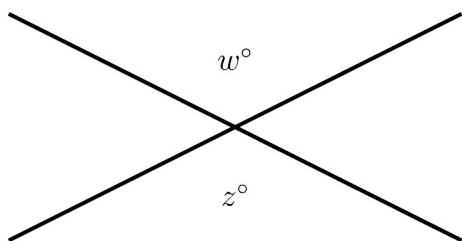
Quadrilaterals $PQRS$ and $WXYZ$ are similar where P, Q , and R correspond to W, X , and Y , respectively. The measure of $\angle S$ is 135° , $PS = 45$, and $WZ = 9$. What is the measure of $\angle Z$?

- A) 5° C) 45°
B) 27° D) 135°

10. [March 2024]

A polygon has exactly 83 sides. If the measures of each of the 83 interior angles of this polygon is $(180p)^\circ$, what value of p ?

11. [March 2024]

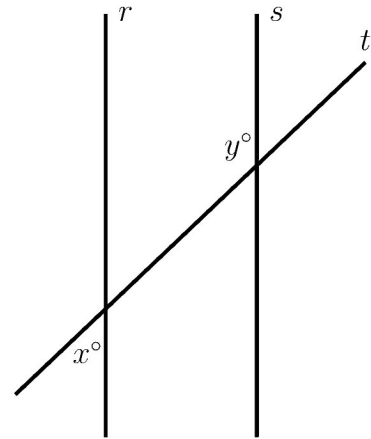


Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. If $w = 150$, what is the value of z ?

- A) 30 C) 75
B) 50 D) 150

12. [May 2024]



Note: Figure not drawn to scale

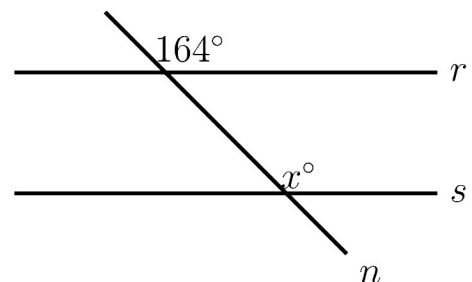
In the figure shown, line r is parallel to line s , and both lines are intersected by line t . If $x = 2w + 24$, $y = 5w + 29$, and $7w = a$, where a is a constant, what is the value of a ?

13. [May US 2024]

Quadrilateral $P'Q'R'S'$ is similar to quadrilateral $PQRS$ where P, Q, R , and S correspond to P', Q', R' , and S' , respectively. The measure of angle P is 45° , the measure of angle Q is 70° , and the measure of angle R is 90° . The length of each side of $P'Q'R'S'$ is 3 times the length of each corresponding side of $PQRS$. What is the measure of angle P' ?

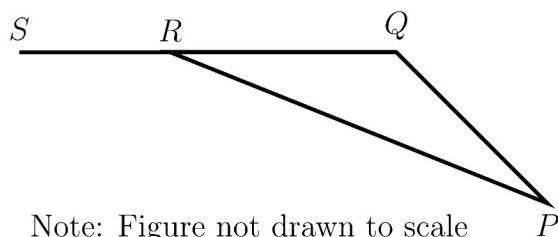
- A) 15° C) 60°
B) 45° D) 135°

14. [June 2024]



In the figure, line n intersects lines r and s . Line r is parallel to line s . What is the value of x ?

15. [June 2024]

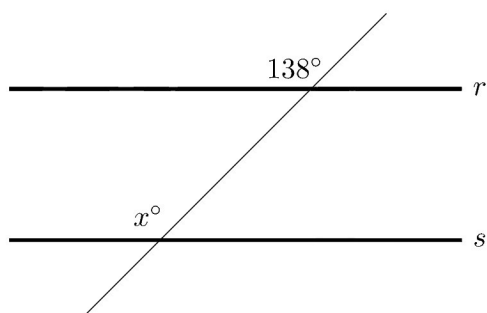


Note: Figure not drawn to scale

In triangle PQR , $\overline{QR}$ is extended to point S . The measure of $\angle PRS$ is 166° . What is the measure of $\angle QPR$?

- A) 14° C) 24°
B) 19° D) 38°

16. [June 2024]



In the figure, line n intersects lines r and s . Line r is parallel to line s . What is the value of x ?

17. [August 2024]

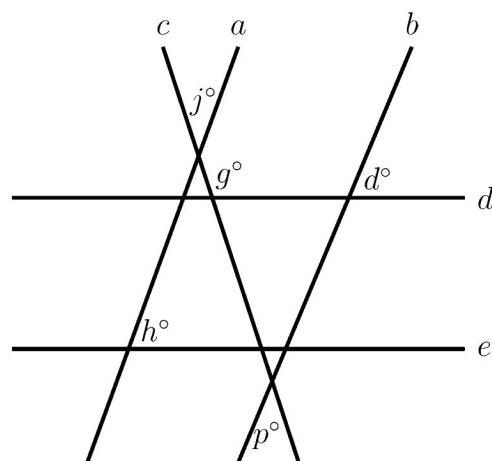
The difference between the measure of angle A and the measure of angle B is $-\frac{3}{4}\pi$ radians. Which expression shows the difference between the measure of angle A and the measure of angle B , in degrees?

- A) $-\frac{3}{4} \cdot \frac{180^\circ}{\pi}$
B) $-\frac{3}{4}\pi \cdot \frac{\pi}{180^\circ}$
C) $-\frac{3}{4}\pi \cdot \frac{180^\circ}{\pi}$
D) $-\frac{3}{4}\pi \cdot \frac{360^\circ}{\pi}$

18. [August US 2024]

In triangle PQR , the measures of angle P is $(3x+5)^\circ$, the measure of angle Q is $(2x+9)^\circ$, and the measure of angle R is $(4y+5)^\circ$. If side QR is extended through point R to point S , and the measure of angle PRS is $(x+y)^\circ$, what is the value of $x+y$?

19. [August 2024]

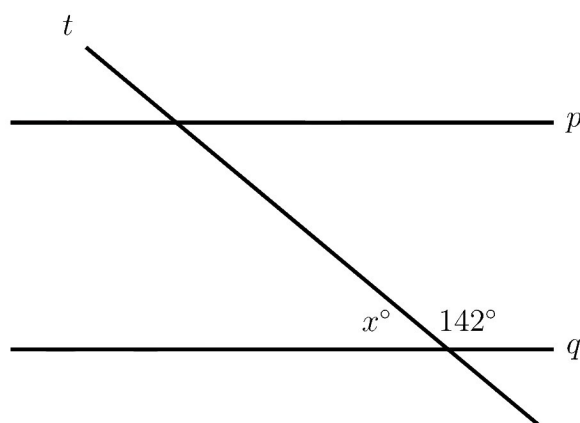


Note: Figure not drawn to scale

In the figure, parallel lines r and s are intersected by lines t , u , and w . If $d = 42$, $g = 143$, and $h > d$, which statement about j and p must be true?

- A) $j = p$ C) $j > p$
B) $j + p = 90$ D) $j < p$

20. [August US 2024]

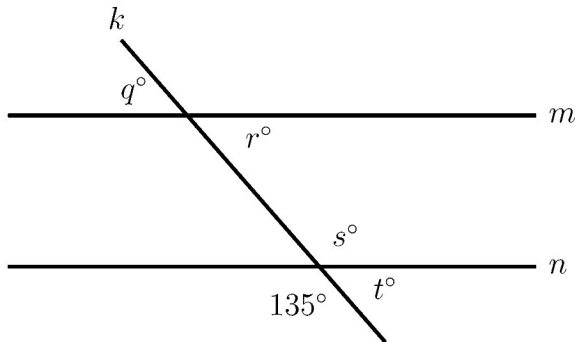


Note: Figure drawn not to scale

In the figure, line p is parallel to line q , and t intersects both of them. What is the value of $x + 142$?

- A) 52 C) 142
B) 90 D) 180

21. [August US 2024]



In the figure shown, line m is parallel to line n , and line k intersects both lines. Which of the following statements is true?

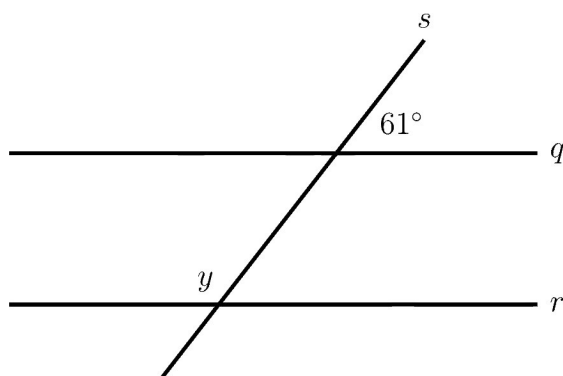
- A) $q = 135$ C) $s = 135$
 B) $r = 135$ D) $t = 135$

22. [October US 2024]

In $\triangle JKL$, the measures of both $\angle J$ and $\angle K$ are equal, and the measure of $\angle L$ is 134° . What is the measure of $\angle J$?

- A) 23° C) 67°
 B) 46° D) 90°

23. [October US 2024]



Note: Figure not drawn to scale

In the figure, line q is parallel to line r , and both lines are intersected by line s . If $y = 2x + 9$, what is the value of x ?

- A) 26 C) 55
 B) 35 D) 65

24. [October 2024]

In triangle ABC , the measure of angle A is 47° , the measure of angle B is 90° , and the measure of angle C is $(\frac{k}{2})^\circ$. What is the value of k ?

- A) 43 C) 86
 B) 45 D) 94

25. [October 2024]

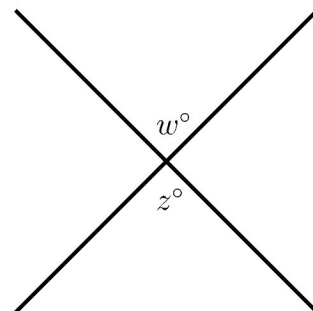
A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $(7x - 250)^\circ$. The sum of the measures of four of the eight angles is k . Which of the following could **NOT** be equivalent to k , for all values of x ?

- A) $-14x + 1,540$ C) $-28x + 1,720$
 B) $14x - 320$ D) 360

26. [October US 2024]

A line intersects two parallel lines, forming four acute and four obtuse angles. The measure of one of the acute angles is $(7x - 619)^\circ$. The sum of the measure of one of the acute and three of the obtuse angles is $(-14x + w)^\circ$. What is the value of w ?

27. [December 2024]

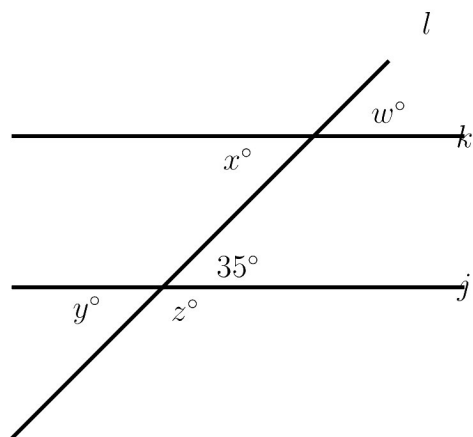


Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. If $w = 128$, what is the value of z ?

- A) 28 C) 64
 B) 52 D) 128

28. [October US 2024]

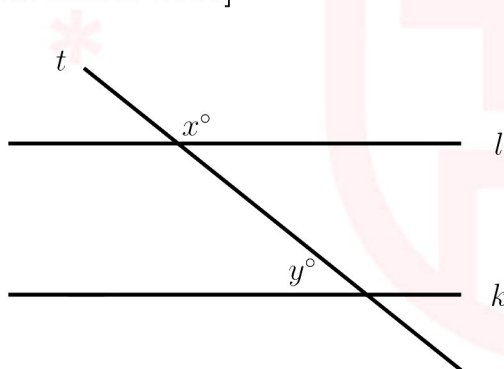


Note: Figure not drawn to scale

In the figure shown, line l intersects lines j and k . Which additional piece of information is sufficient to prove that lines j and k are parallel?

- A) $w = 35^\circ$
- B) $x = 145^\circ$
- C) $y = 35^\circ$
- D) $z = 145^\circ$

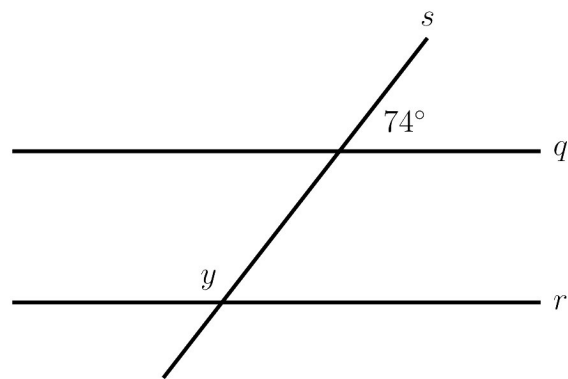
29. [November 2024]



In the figure shown, the lines l and k are parallel and line t intersects both lines. If $x > 154$, which of the following must be true?

- A) $y < 26$
- B) $y > 26$
- C) $x + y < 180$
- D) $x + y > 180$

30. [December 2024]



Note: Figure not drawn to scale

In the figure, line q is parallel to line r , and both lines are intersected by line s . If $y = 2x + 8$, what is the value of x ?

- A) 33
- B) 41
- C) 49
- D) 59